

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Code of Conduct:

Abangjithan S /

I 192521299 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

S. Abangjithan

Experiments

QUESTIONS:4

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP. (BL4) (10)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1. TCP Three-way handshake

Packets involved :

1. SYN (synchronize)

→ sent by the client to initiate a connection.

→ SYN flag = 1

→ contains the client's initial sequence number (ISN).

2. SYN-ACK (synchronize - Acknowledge)

→ Sent by the server in response.

→ SYN = 1, ACK = 1.

→ Acknowledges the client's sequence number and sends the server's ISN.

3. ACK (Acknowledgment)

→ Sent by the client.

→ ACK flag = 1

→ Acknowledges the server's sequence number.

Role of each flag :

→ SYN : starts the TCP connection and synchronizes sequence numbers.

→ ACK: confirms successful receipt of packets.

→ SYN-ACK: combines synchronization and acknowledgment in one packet.

How reliability is established:

→ Both devices exchange and acknowledge sequence numbers.

→ Ensures both sender and receiver are ready before data transfer begins.

→ Prevents data loss and duplicate connections.

2. DNS query using UDP

Comparison of UDP and TCP headers.

Feature	UDP	TCP
Header size	8 bytes	20-60 bytes
Sequence number	No	Yes
Acknowledgement	No	Yes
connection	connections	connection-oriented
reliability	No	Yes

Impact of UDP lightweight design :

- Smaller header reduces overhead.
- Faster transmission.
- No connection setup required.
- No retransmission or acknowledgment.
- Suitable for DNS because queries are small and quick.
- If a packet is lost, the application simply sends another request.

3. TCP vs UDP During Packet Loss

Observation in Wireshark :

TCP :

- Detects Packet loss.
- Retransmits lost packets.
- Uses acknowledgments, sequence numbers and timeout mechanisms.

UDP :

- Does not detect packet loss.
- No retransmission.
- Lost packets are discarded permanently.

Why TCP is reliable but slower :

→ Error detection and recovery.

→ Flow control.

→ congestion control.

→ Retransmissions increase delay.

Why UDP is faster:

→ No connection establishment.

→ No acknowledgments.

→ No retransmissions overhead.

→ Suitable for real-time applications

like video streaming, VoIP, and online gaming.

4. DNS Resolution using Wireshark.

Typical DNS query:

→ Domain: www.google.com

→ Query type: A

→ Request: client asks for the IPv4 address.

Response:

→ DNS server returns the IPv4 address. (eg: 142.250.xxx.xxx)

other query types:

- A : IPv4 Address.
- AAA : IPv6 Address.
- MX : Mail server record.

Time taken for DNS resolution:

- usually 5-50 ms on a local network.
- can vary depending on network conditions and DNS server.

Why DNS uses UDP:

- Fast and efficient.
- ~~Small~~ packet size.
- No connection setup.
- Low protocol overhead.
- If a response is too large or truncated, DNS switches to TCP.